

1. Course number and name

CHE 433. Process Simulation with Aspen Plus

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Process Simulation with Aspen Plus Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020.

a. other supplemental materials

- Ralph Schefflan, Teach yourself the Basics of Aspen Plus, 2nd Edition, Wiley. (2016).
- Gavin Towler and Ray Sinnott, Chemical Engineering Design, 2nd Ed, Elsevier Ltd. (2013).
- Lorenz T. Biegler, Ignacio E. Grossmann and Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International Series, December (1999).
- T. F. Edgar and D. M. Himmelblau, Optimization of Chemical Processes, McGraw-Hill, Inc. (1988).
- A. W. Westerberg, H. P. Hutchison, R. L. Motard and P. Winter, Process Flowsheeting, 1st Edition, Cambridge University Press (1979).
- SOFTWARE: Athena Visual Studio from www.AthenaVisual.com; EXCEL, ASPEN Plus and optionally MATLAB and ASPEN Hysys.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Main principles of Process Simulation (Material and Energy Balances) relevant to chemical and biological processes. Process Simulation or Flowsheeting is defined as the use of commercially available products (like ASPEN Plus, ASPEN Hysys, UniSim, VMG, ChemCad, gProms, Pro II) to perform steady-state Heat and Weight Balances, Sizing and Cost calculations for chemical, biological, petrochemical and refining processes.

b. Prerequisites or co-requisites

CHE 312 and CHE 313 and CHE 321; or consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Select and justify choices for physical property models required for accurate heat and weight balance calculations.
- Design and model unit operations such as chemical reactors, heat exchangers, heaters and coolers, pumps, compressors and turbines, mixers, distillation and packed columns and user specific operations.
- Combine units operations in a cohesive Flowsheet that describes existing or conceptual processes.
- Familiarize themselves with computational aspects such as design specs, calculator blocks, sensitivity analyses, flow sheet convergence

- Perform optimization studies.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	

7. **Brief list of topics to be covered**

INTRODUCTION TO PROCESS FLOWSHEETING IN CHEMICAL ENGINEERING: Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Process Simulation, Tutorial on Programming with ASPEN Plus and Athena Visual Studio

THERMODYNAMICS AND PROCESS UNIT OPERATIONS: Vapor-Liquid Equilibrium and Property Methods, VLE Data Regression, Mixers and Splitters, Two and Three Phase Flash Blocks, Heaters, Coolers and Condensers, Distillation, Pumps, Compressors and Turbines, Example Flowsheets and Design Specifications

PROCESS FLOWSHEETING FUNDAMENTALS: Recycle Stream Selection and Tear Equations, Convergence Methods for Tear Equations, Calculator and Transfer Function Blocks, Simultaneous Recycle Convergence and Design Specifications, Sensitivity Analysis and Case Studies, Sequential Modular vs. Equation Based Solution Methods

CHEMICAL KINETICS AND REACTOR UNIT OPERATIONS CSTR: Plug Flow and Fixed Bed Reactors, Conversion and Equilibrium Reactors, Example Flowsheets with Design Specifications, Excel Interface for Custom Reactor and Physical Models

COMPLEX UNIT OPERATIONS AND ADVANCED SIMULATION TOPICS: Packed columns for Absorption and Stripping Operations, Rate based vs. Equilibrium Models, Homogeneous and Heterogeneous Azeotropic Distillation, Process Optimization and Pinch Analysis for Energy Optimization, Heat Exchangers and Heat Exchanger Network Optimization, Solids Handling and Electrolyte Systems